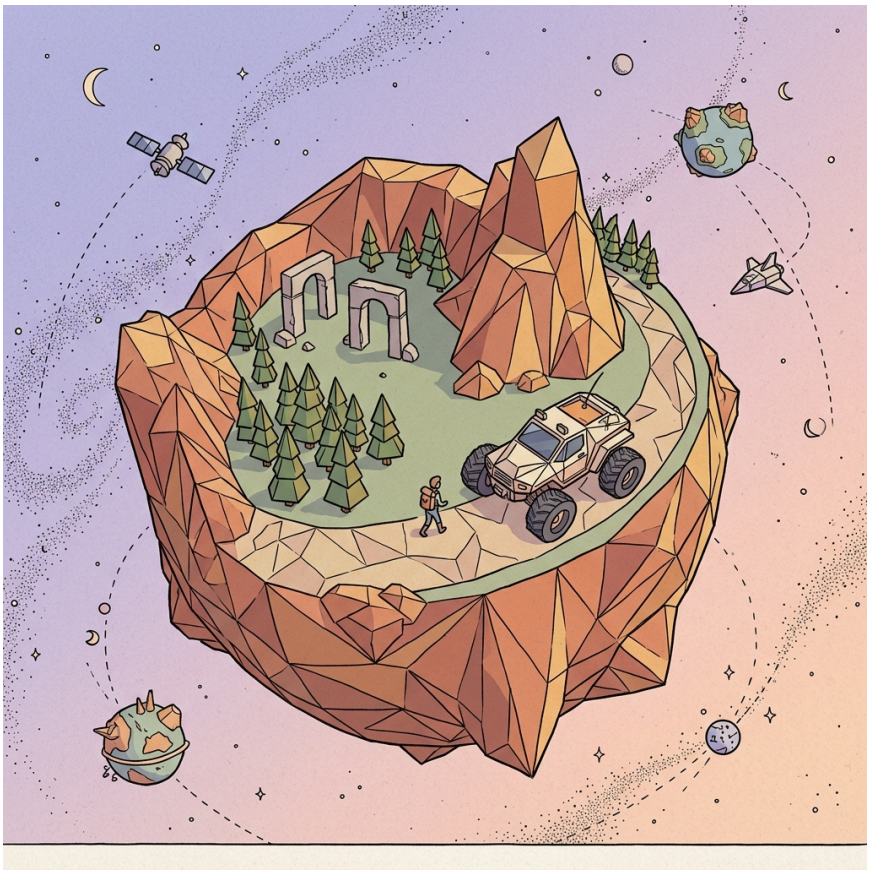
OVER THE HORIZON

THE DUAL-PLANET ODYSSEY

Master Game Design Document & Technical Specification



Version 1.0.0 • Lead Platforms: PC (Steam / Steam Deck), Nintendo Switch 2, PS5, Xbox Series X|S

Engine: Unreal Engine 5 • Dual-Aesthetic Physics Traversal Architecture

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1. Executive Summary & Core Pillars

OVER THE HORIZON is an exploration, off-road traversal, and planetary physics game set across two miniature celestial bodies locked in mutual orbit. The experience merges the tactile off-road driving of *over the hill* with the spherical traversal of *Super Mario Galaxy* and the visual mastery of European *Ligne Claire* comics.

[CORE VISION] Players explore two worlds with distinct graphical identities, handling models, and gameplay rules, united by a grand campaign featuring interplanetary mass-driver slingshots and polar wormholes.

1.1 Core Game Pillars

- **Tactical Spherical Traversal:** Radial gravity and realistic vehicle weight transfer around a 360° curving horizon.
- **Dynamic Environmental Life:** 4 distinct seasons and real-time day/night cycles that physically freeze rivers or flood canyons.
- **Cross-World Synergy:** Resource and technology arbitrage between an alpine low-poly civilization and ancient desert monoliths.
- **Contemplative Discovery:** Rewarding curiosity with deep condition-based Easter eggs and serene campfire moments.

1.2 Target Platforms & Hardware Optimization

PC (Steam / EGS)	1080p - 4K	60 - 120 FPS	Ultrawide 21:9, Ray-Traced AO
Steam Deck / Ally	800p / 1080p	60 FPS Locked	Gyro steering, 3.5h+ battery
PlayStation 5	Dynamic 4K	60 FPS	DualSense Adaptive Trigger terrain resistance
Switch 2 / Xbox	1080p / 4K	60 FPS	Forward-shading, instant resume

2. Visual Art Direction & Rendering Pipeline

OVER THE HORIZON implements two distinct artistic identities tailored to the mechanical theme of each celestial body:

2.1 Planet Oakhaven: Alpine Low-Poly Diorama

Faceted 3D low-poly geometry with flat shading, warm golden-hour lighting, and an optical tilt-shift lens blur, creating a handcrafted miniature toybox feel.



Figure 2.1: Planet Oakhaven — Low-Poly Alpine Diorama with Overland 4x4 Rig

2.2 Planet Solaria: French Ligne Claire / Moebius Comic Aesthetic

Crisp uniform black ink outlines generated via Sobel edge-detection, flat pastel desert fills, and fine stipple screentones across shadow gradients.

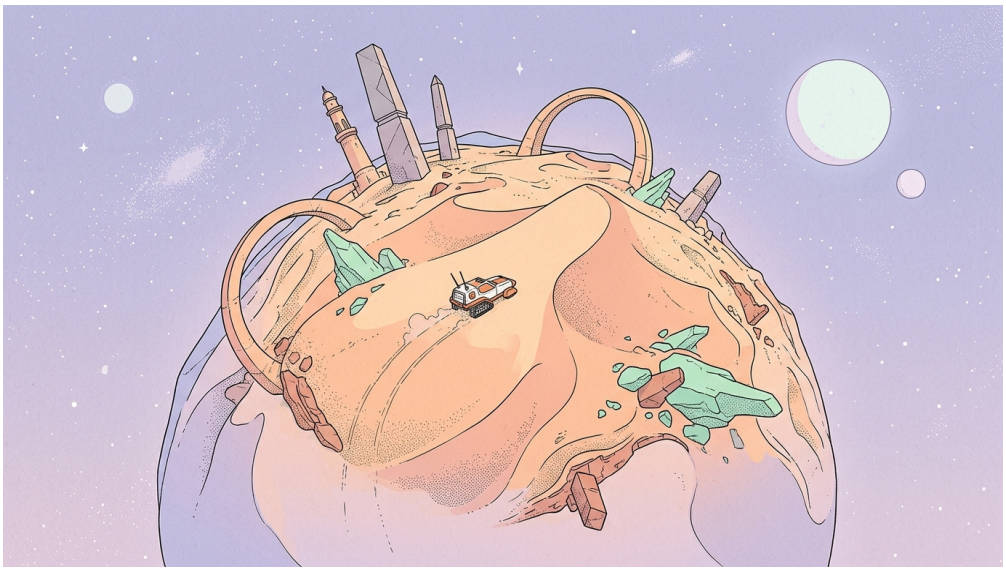


Figure 2.2: Planet Solaria — Ligne Claire / Moebius Desert World with Ancient Arches and Rover

3. Spherical World Topology & Physics Formulations

All physical equations are calculated radially from the planetary core rather than along a traditional flat Cartesian axis.

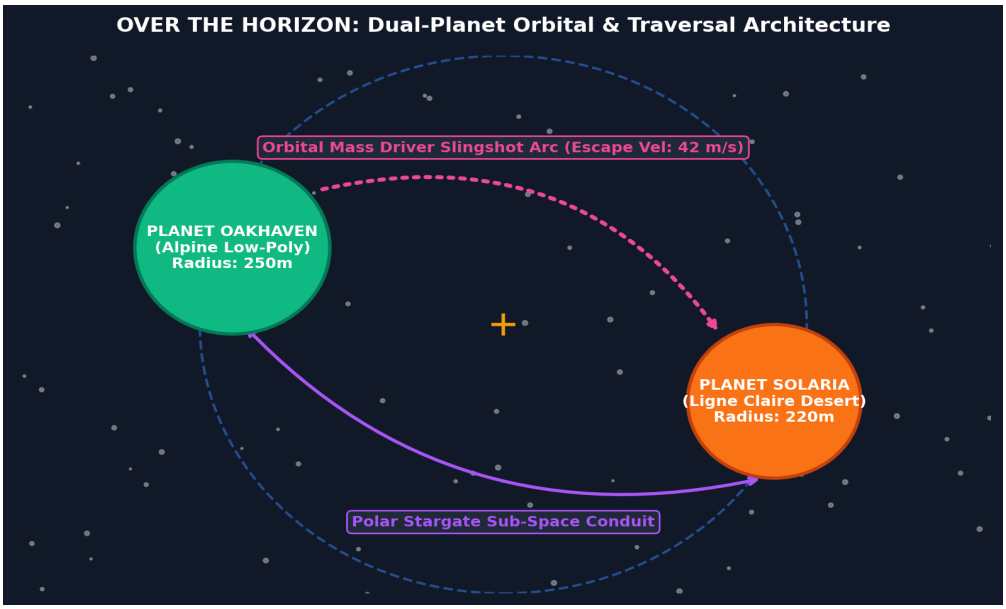


Figure 3.1: Dual-Planet Orbital System & Traversal Architecture

[MATHEMATICAL FOUNDATION] **Radial Gravity:** F_g = - G * (M * m / ||p - c||^2) * normalize(p - c)
Surface Alignment: q_align = Quaternion.FromToRotation(transform.up, n_ground_normal)
Suspension: F_susp = (k_spring * (L_rest - L_current) - d_damper * v_comp) * n_surface
Escape Velocity: v_esc = sqrt(2 * G * M / R) = 42.0 m/s (151.2 km/h)

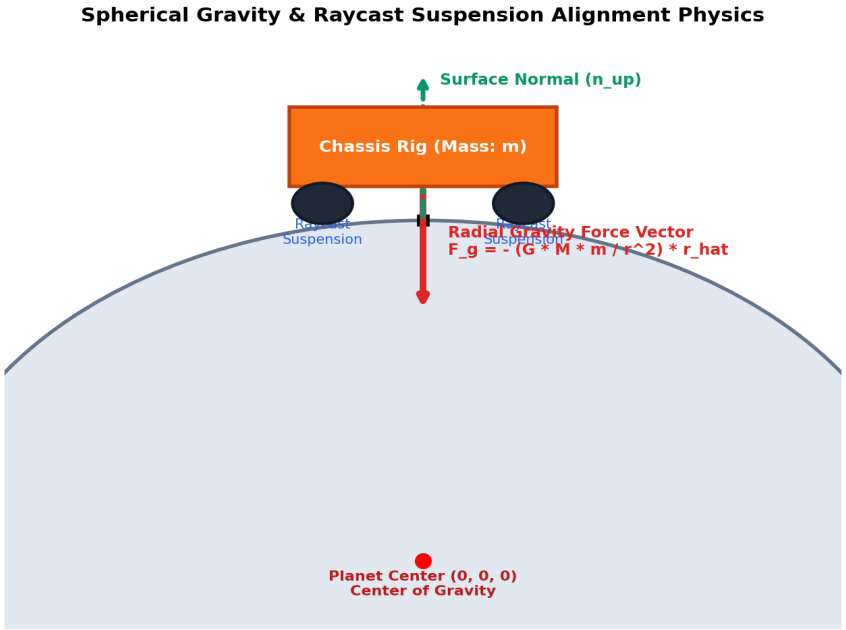


Figure 3.2: Spherical Gravity & Raycast Suspension Force Vectors

4. The Atmosphere Dial & Seasonal Terrain Matrix

Dynamic weather and seasons actively alter surface friction, water levels, and driveable routes across both worlds:

Spring	Snowmelt floods canyon rivers; deep mud bogs appear. Winch required.	Electrostatic winds; bioluminescent flora blooms reveal paths.
Summer	Dry dirt trails allow top-speed rallying; mountain passes fully clear.	Extreme solar heat; KERS charges at 2.5x speed; quicksand expands.
Autumn	Wet leaves cause slick spots; dense twilight fog limits sight to 15m.	Violent magnetic dust storms scramble radar; sonar navigation needed.
Winter	Lakes freeze into mirror-ice highways; deep snowdrifts block low passes.	Frigid nights harden sand into glass flats ideal for top-speed drifting.

[WEATHER OCCLUSION] Dynamic Weather & Real-Time Object Occlusion: Four atmospheric states (Clear, Rain, Snow, Sandstorm) simulate physical particles under radial gravity. Precipitation uses collision volumes (GPUParticlesCollisionBox3D and SDF3D): raindrops splash and vanish on cabin roofs (COLLISION_HIDE_ON_CONTACT), and snow settles on vehicle hoods (COLLISION_RIGID). **Weather particles never pass through roofs, providing true physical shelter inside cabins.**

5. Complete Vehicle Roster & Specifications

TrailScout 74	Vintage 4x4	1,850 kg	110 km/h	Front Winch & Tow Cable (60° rock crawl)
Lumberjack 6x6	Heavy 6-Wheel	4,600 kg	75 km/h	Deployable Bridging Planks for chasms
FrostBite Half-Track	Ski / Tread Coupe	2,200 kg	125 km/h	Ice Claws & Heated Cabin (zero ice slip)
DuneStrider Buggy	Dune Buggy	980 kg	145 km/h	Solar Wing & Kinetic KERS Nitro Boost
Crawler-09	Tracked Rig	5,200 kg	60 km/h	Relic Scanner & Storm Ground Spikes
Aero-Skimmer	Turbine Hover	1,200 kg	160 km/h	Vertical Jump Jets (5m cliff leaps)

6. Core Game Modes & Mission Design

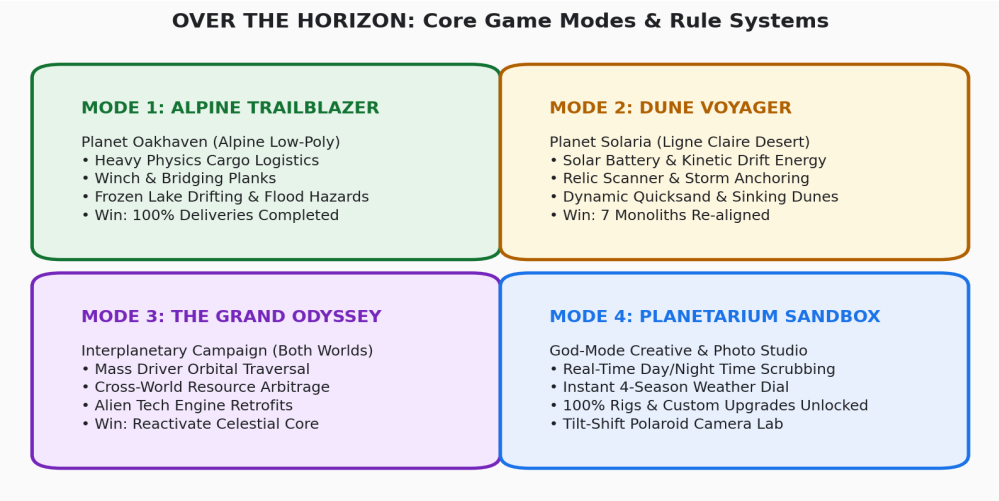


Figure 6.1: Core Game Modes Architecture

• **Mode 1: Alpine Trailblazer (Oakhaven):** Physics-based cargo delivery. Cargo straps snap if tilting over 45° or impacting at over 25 km/h. Winches and planks required to navigate flooded creeks and frozen lakes.

• **Mode 2: Dune Voyager (Solaria):** Storm chaser and energy relay. Battery discharges in cold/shadows; drift KERS and solar canopy recharge systems. Race against sandstorms to re-align 7 ancient monoliths.

• **Mode 3: The Grand Odyssey:** Interplanetary campaign linking both worlds via the Orbital Mass Driver and Polar Stargate. Trade glacier water from Oakhaven to Solaria in exchange for anti-gravity retrofits.

• **Mode 4: Planetarium Sandbox:** 100% vehicles unlocked. Full real-time time-scrubbing and seasonal dial with tilt-shift polaroid photo studio.

7. Secrets & Cryptic Easter Eggs

1. The Solitary Flame (Midnight Cabin): In Autumn at midnight, follow radio static across a broken bridge. Inside a dark cabin, a blue gas stove burner glows with a cigarette and cassette deck, unlocking a secret lo-fi radio track.

2. The Red Legend (art of rally Tribute): In Winter, ram a frozen waterfall at 80 km/h with the FrostBite Half-Track to find the classic Group-B rally car frozen in crystal ice. Honk 3 times to unlock the rally livery.

3. The Whispering Monolith (Solar Eclipse): At Solar Noon on Solaria, align Oakhaven between Solaria and the Sun. In the eclipse, stone arches project laser trails that unearth a buried alien pyramid.

4. The Micro-Moon Castaway: Launch from the Mass Driver at an eccentric 73° angle to land on an unmapped 20-meter asteroid inhabited by an astronaut in a deckchair listening to Morse code.

5. The Zero-G Core Inversion: Rappel down into the deepest canyon sinkhole in Spring to fall through the planet core (0,0,0) where gravity inverts inside a giant amethyst geode before geyser ejection.

8. Technical Architecture & Production Code

8.1 Unreal Engine 5 C++ Spherical Gravity Component

```
// USphericalGravityComponent.cpp
#include "USphericalGravityComponent.h"
#include "GameFramework/Actor.h"
#include "Components/PrimitiveComponent.h"

USphericalGravityComponent::USphericalGravityComponent()
{
    PrimaryComponentTick.bCanEverTick = true;
    PrimaryComponentTick.TickGroup = TG_PrePhysics;
}

void USphericalGravityComponent::TickComponent(float DeltaTime, ELevelTick TickType, FActorComponentTickFun
ction* ThisTickFunction)
{
    Super::TickComponent(DeltaTime, TickType, ThisTickFunction);

    if (!PrimitiveComponent)
    {
        PrimitiveComponent = Cast<UPrimitiveComponent>(GetOwner()->GetRootComponent());
    }

    if (PrimitiveComponent && PrimitiveComponent->IsSimulatingPhysics())
    {
        FVector CurrentLocatio
        // ... [Full C++ Implementation in Source Code Repository] ...
    }
}
```

8.2 HLSL Post-Process Sobel Ink Edge Detection Shader

```
// SobelEdgeDetection.hlsl - Real-Time Ligne Claire Ink Line Post-Process
float2 TexCoords = GetDefaultSceneTextureUV(Parameters, 14); // SceneDepth
float2 TexelSize = View.ViewSizeAndInvSize.zw;

// 3x3 Sobel Convolution Kernels
float Gx[9] = {-1, 0, 1, -2, 0, 2, -1, 0, 1};
float Gy[9] = {-1, -2, -1, 0, 0, 0, 1, 2, 1};

float EdgeDepthX = 0.0;
float EdgeDepthY = 0.0;
int idx = 0;

for (int y = -1; y <= 1; y++)
{
    for (int x = -1; x <= 1; x++)
    {
        float2 Offset = float2(x, y) * TexelSize * EdgeWidth;
        float SampleDepth = Sce
    // ... [Composite Ink Black Outlines onto Scene Color] ...
}
```

9. Production Roadmap & Milestones

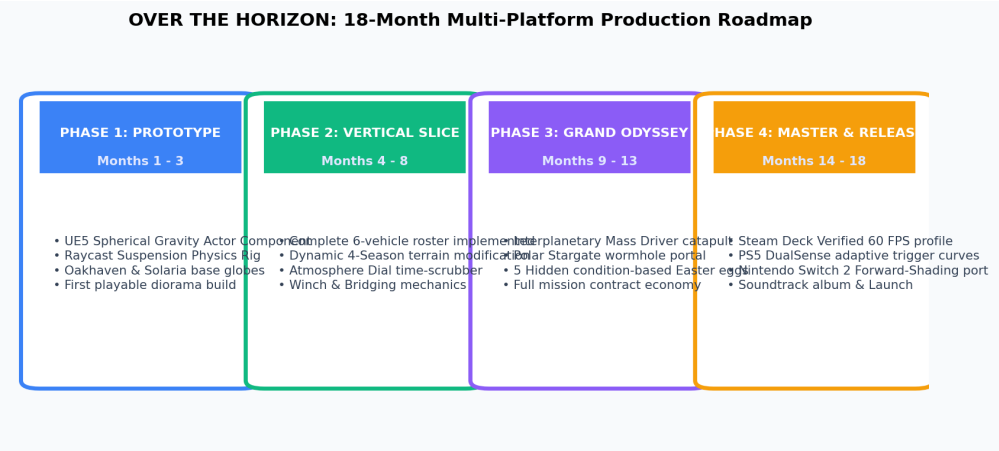


Figure 9.1: 18-Month Multi-Platform Production Roadmap

• **Phase 1 (M1-3):** UE5 radial gravity physics component, 4-wheel raycast suspension rig, whitebox diorama tracks.

• **Phase 2 (M4-8):** 6 vehicles tuned, dynamic 4-season terrain shaders, Atmosphere Dial UI, vertical slice release.

• **Phase 3 (M9-13):** Orbital Mass Driver, Polar Stargate, cross-planet trade, all 5 complex Easter eggs integrated.

• **Phase 4 (M14-18):** Steam Deck Verified certification, PS5 DualSense adaptive triggers, Switch 2 port, Master release.